

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 3, all 12 PYQ S-parameter problems, 2069–2082 BS

3. RF and M/W Network Theory and Analysis: Numerical Problems

22 numerical marks, 1.1% of the archive, 17% of this chapter · 3 solved PYQ problems · across 3 papers · 2 numeric matrices, 1 modulator

READ THIS FIRST

Chapter 3 numericals are not arithmetic problems. Every one of them is the same four tests of §3.3 applied to a matrix. **Nothing here needs a calculator beyond a square and a log.** Every number below was computed and asserted in `src/sparam.py`.

- **Always run the four tests in the same order, and always write them as a table.** Examiners give a mark per test; a paragraph hides them.
- **A matrix that is not unitary is not wrong**, it is lossy. Say so, and say how much: the shortfall in a column is the dissipated fraction.
- **The tee problems live in Chapter 4.** The E-plane, H-plane and magic-tee derivations, the three device identifications and the two radar problems are filed with the tee junctions, so they are the **Chapter 4 companion, Problems 6 to 17**. They use the same four tests.

M. The Four Tests, and the Conversions

M1: the four tests (Problems 1 and 2; Ch4 companion Problems 13 to 15)

1. **Diagonal:** which $S_{ii} = 0$? Those ports are matched.
2. **Transpose:** is $S_{ij} = S_{ji}$ for every pair? Yes means reciprocal (passive, isotropic); no means ferrite or active.
3. **Column sums:** is $\sum_k |S_{ki}|^2 = 1$ for every column? Yes means lossless; the shortfall is the dissipated fraction of the incident power.
4. **Column dots:** is $\sum_k S_{ki} S_{kj}^* = 0$ for every $i \neq j$? This is the phase half of the unitary condition.

Always write M1 as this table

Test	Computation	Conclusion
Diagonal	every S_{ii}	matched / not
Transpose	S_{ij} vs S_{ji}	reciprocal / not
Column sums	$\sum_k S_{ki} ^2$	lossless, or lossy by the shortfall
Column dots	$\sum_k S_{ki} S_{kj}^*$	phases consistent / not

The conversions used throughout

- $RL = -20 \log_{10} |S_{11}|$ dB, $IL = -20 \log_{10} |S_{21}|$ dB.
- $G = +20 \log_{10} |S_{21}|$ dB, $VSWR = \frac{1 + |S_{11}|}{1 - |S_{11}|}$.
- $P_{\text{ref}} = |S_{11}|^2 P_{\text{inc}}$, $P_{\text{trans}} = |S_{21}|^2 P_{\text{inc}}$.
- $P_{\text{diss}} = (1 - \sum_k |S_{k1}|^2) P_{\text{inc}}$.

Power bookkeeping (Problem 1)

1. Write the drive vector $[a]$; multiply, $[b] = [S][a]$.
2. Power leaving port i is $|b_i|^2$, in the same units as $|a|^2$.
3. Check the books: $\sum_i |b_i|^2 \leq \sum_j |a_j|^2$, with equality for a lossless network.

Problem 1: The four tests on a numeric two-port matrix 3+1+1+1+2

79 Bh · the only fully numeric S-matrix in the archive

QUESTION AS PRINTED 79 Bh [3+1+1+1+2]

Why are S-parameters used in high frequencies? The S-matrix of a certain microwave network is given as $[S] = \begin{bmatrix} 0.4 + j0.5 & j0.6 \\ j0.6 & 0.4 - j0.5 \end{bmatrix}$.

(a) Is the network reciprocal? (b) Is the network lossless? (c) What is the return loss at the input? (d) If the input power to the network is 5 watts, what is the reflected power?

Given

- $S_{11} = 0.4 + j0.5$, $S_{12} = S_{21} = j0.6$, $S_{22} = 0.4 - j0.5$.
- Input power $P_{in} = 5$ W at port 1; port 2 terminated in Z_0 .

Step 0: the magnitudes, computed once and reused

- $|S_{11}| = \sqrt{0.4^2 + 0.5^2} = \sqrt{0.16 + 0.25} = \sqrt{0.41} = 0.6403$.
- $|S_{22}| = \sqrt{0.4^2 + (-0.5)^2} = \sqrt{0.41} = 0.6403$ (the same).
- $|S_{12}| = |S_{21}| = |j0.6| = 0.6$.
- Squares, which is what every power test needs: $|S_{11}|^2 = |S_{22}|^2 = 0.41$, $|S_{12}|^2 = |S_{21}|^2 = 0.36$.

Step 1: run the four tests

Test	Computation	Result	Conclusion
Diagonal	$S_{11} = 0.4 + j0.5 \neq 0$, $S_{22} = 0.4 - j0.5 \neq 0$	both non-zero	Neither port is matched
Transpose	$S_{12} = j0.6$ and $S_{21} = j0.6$	$S_{12} = S_{21}$	(a) Reciprocal ✓
Column 1 sum	$ S_{11} ^2 + S_{21} ^2 = 0.41 + 0.36$	$0.77 \neq 1$	(b) Not lossless
Column 2 sum	$ S_{12} ^2 + S_{22} ^2 = 0.36 + 0.41$	$0.77 \neq 1$	Same shortfall at port 2
Column dot	$S_{11}S_{12}^* + S_{21}S_{22}^* = (0.4 + j0.5)(-j0.6) + (j0.6)(0.4 + j0.5)$ $= (0.30 - j0.24) + (-0.30 + j0.24)$	0	Phases are consistent; only the magnitudes fail

- **(a)** $S_{12} = S_{21} = j0.6$, so the network is **reciprocal**: it is passive and contains no ferrite and no active device.
- **(b)** Both columns sum to 0.77, not 1, so the network is **not lossless**. It **dissipates** $1 - 0.77 = 0.23$, i.e. **23%** of whatever is fed into either port.
- Worth adding: the column-dot test *passes*. A network can have the right phase relations and still be lossy, which is why both halves of the unitary condition must be checked.

Step 2: (c) return loss at the input

- Return loss compares incident with reflected power at port 1, and S_{11} is an **amplitude** ratio, so the factor is 20:

$$RL = -20 \log_{10} |S_{11}| = -20 \log_{10}(0.6403)$$

- $\log_{10}(0.6403) = -0.19361$, so $RL = -20 \times (-0.19361)$.

- **(c)** **$RL = 3.87$ dB**.

- Sanity check: a low return loss means a **bad** match, and indeed $VSWR = \frac{1 + 0.6403}{1 - 0.6403} = \frac{1.6403}{0.3597} = 4.56$, far from 1.

Step 3: (d) reflected power for 5 W in

- Power scales as the **square** of the wave amplitude:

$$P_{ref} = |S_{11}|^2 P_{in} = 0.41 \times 5$$

- **(d)** **$P_{ref} = 2.05$ W**.

Step 4: complete the power budget (free marks)

Component	Formula	Value	Share of 5 W
Reflected at port 1	$ S_{11} ^2 P_{in} = 0.41 \times 5$	2.05 W	41%
Delivered to port 2	$ S_{21} ^2 P_{in} = 0.36 \times 5$	1.80 W	36%
Dissipated inside	$(1 - 0.77) \times 5$	1.15 W	23%
Total		5.00 W	100%

- The three rows sum to the input, which is the arithmetic check that the whole answer is consistent.
- Insertion loss, if asked: $IL = -20 \log_{10}(0.6) = \mathbf{4.44}$ dB.

Answer to the theory rider (3 marks)

- S-parameters are used at high frequencies because **total V and I are not uniquely defined** when the circuit is comparable to λ , because **open and short terminations cannot be realised** broadband and would **destroy or oscillate** active devices, and because **power** is what microwave instruments actually measure. See §3.1.

Problem 2: Comparing two amplifier S-parameter sets 4

82 Bh · $Z_0 = 50\ \Omega$; a reading question, not a design question

QUESTION AS PRINTED 82 Bh [4]

Find the basic differences of the systems having the following sets of S-matrix.

- (a) $S_{11} = 0.65\angle 146^\circ$, $S_{12} = 0.12\angle 46^\circ$, $S_{21} = 2.30\angle 44^\circ$, $S_{22} = 0.17\angle -172^\circ$
 (b) $S_{11} = 0.30\angle -134^\circ$, $S_{12} = 0.03\angle 46^\circ$, $S_{21} = 2.20\angle 44^\circ$, $S_{22} = 0.10\angle -172^\circ$

Given

- Two **two-port** sets, quoted in polar form. Only the **magnitudes** matter for the comparison the question asks for; the angles matter only for Δ and K .

Step 1: what kind of device is each? (the two tests that decide it)

Test	Computation	Result	Conclusion
Transpose	(a) $ S_{12} = 0.12$ vs $ S_{21} = 2.30$; (b) 0.03 vs 2.20	$S_{12} \neq S_{21}$ in both	Both are non-reciprocal
Column 1 sum	(a) $0.65^2 + 2.30^2 = 0.4225 + 5.29$; (b) $0.30^2 + 2.20^2$	$\gg 1$	Both are active: they add power

- $|S_{21}| > 1$ means the device **has gain**, so neither set can describe a passive component. Both are **transistor amplifiers** at the same frequency; the question is which is the better one.

Step 2: convert every magnitude into its engineering figure

Quantity	Formula	System (a)	System (b)	Better
Forward gain	$20 \log_{10} S_{21} $	7.23 dB	6.85 dB	(a), barely
Reverse isolation	$-20 \log_{10} S_{12} $	18.42 dB	30.46 dB	(b), by 12 dB
Input return loss	$-20 \log_{10} S_{11} $	3.74 dB	10.46 dB	(b)
Input VSWR	$\frac{1+ S_{11} }{1- S_{11} }$	4.71	1.86	(b)
Output return loss	$-20 \log_{10} S_{22} $	15.39 dB	20.00 dB	(b)
Output VSWR	$\frac{1+ S_{22} }{1- S_{22} }$	1.41	1.22	(b)

- Sample line, so the arithmetic is visible: $-20 \log_{10}(0.65) = -20 \times (-0.18709) = 3.74$ dB, and $\text{VSWR} = (1 + 0.65)/(1 - 0.65) = 1.65/0.35 = 4.71$.

Step 3: stability, the one place the angles are needed

- $\Delta = S_{11}S_{22} - S_{12}S_{21}$, and $K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$.
- **System (a):**
 - $S_{11}S_{22} = (0.65)(0.17)\angle(146^\circ - 172^\circ) = 0.1105\angle -26^\circ = 0.0993 - j0.0484$.
 - $S_{12}S_{21} = (0.12)(2.30)\angle(46^\circ + 44^\circ) = 0.276\angle 90^\circ = j0.276$.
 - $\Delta = 0.0993 - j0.0484 - j0.276 = 0.0993 - j0.3244$, so $|\Delta| = 0.3393$.
 - $K = \frac{1 - 0.4225 - 0.0289 + 0.1151}{2 \times 0.276} = \frac{0.6637}{0.552} = 1.202$.
- **System (b):**
 - $S_{11}S_{22} = (0.30)(0.10)\angle(-134^\circ - 172^\circ) = 0.03\angle 54^\circ = 0.0176 + j0.0243$.
 - $S_{12}S_{21} = (0.03)(2.20)\angle 90^\circ = j0.066$.
 - $\Delta = 0.0176 - j0.0417$, so $|\Delta| = 0.0453$.
 - $K = \frac{1 - 0.09 - 0.01 + 0.0021}{2 \times 0.066} = \frac{0.9021}{0.132} = 6.834$.
- Both have $K > 1$ and $|\Delta| < 1$, so both are **unconditionally stable**, but (b) is stable by a wide margin and (a) only just.

Answer: the basic differences

- Both are **active, non-reciprocal two-ports**, i.e. amplifiers of nearly the same gain (7.23 dB against 6.85 dB, a difference of 0.4 dB that is within normal device spread).
- **The real difference is everywhere except the gain:**
 - **(b) is far better matched at the input:** RL 10.46 dB against 3.74 dB, VSWR 1.86 against 4.71. System (a) reflects 42% of the drive power straight back at the source.
 - **(b) is far better isolated:** $|S_{12}| = 0.03$ against 0.12, an extra 12 dB. System (b) is close to **unilateral**, so it can be designed with the simple unilateral equations; system (a) needs the full bilateral treatment.
 - **(b) is more stable:** $K = 6.83$ against 1.20, and $|\Delta| = 0.045$ against 0.339. System (a) sits near the edge of unconditional stability, so a mismatched load could push it into oscillation.
- **Verdict:** (b) is the better amplifier at a cost of 0.4 dB of gain: better match, better isolation, much better stability margin.
- **Note for Chapter 6:** the same paper then asks for the maximum unilateral and bilateral gains of set (a). For the record they are $G_{TU,\max} = 9.75$ dB and $G_{\max} = 10.11$ dB; the derivation belongs to the amplifier chapter.

Problem 3: Evaluate the S-matrix of an amplitude modulator 778 Ch · verified/added no source note carries this; modelled and evaluated here**QUESTION AS PRINTED** 78 Ch [3+7]

Elaborate the importance of S-parameters for microwave frequencies. Evaluate the S-matrix of an Amplitude Modulator.

Step 1: model the modulator as a two-port

- A microwave amplitude modulator sits **in** the RF path: the carrier enters port 1 and the modulated carrier leaves port 2. The modulating signal is a **control input**, applied as bias, not as a third RF port.
- Physically it is a **PIN-diode variable attenuator**: the diode's RF resistance is set by its DC bias, so the modulating waveform steers the transmission through the device.
- Both RF ports are designed to stay **matched** across the modulation swing, otherwise the modulation would pull the source and produce incidental phase modulation.

Step 2: impose the three properties

Property	Equation	Why
Matched at both ports	$S_{11} = S_{22} = 0$	The attenuator is built as a matched pad at every bias point
Reciprocal	$S_{12} = S_{21} = \alpha$	Passive, no ferrite and no gain: a diode resistance is bilateral
Lossy, and time varying	$0 \leq \alpha(t) \leq 1$	The power removed from the carrier is dissipated in the diode

- Hence

$$[S] = \begin{bmatrix} 0 & \alpha(t) \\ \alpha(t) & 0 \end{bmatrix}, \quad \alpha(t) = \alpha_0 (1 + m \cos \omega_m t)$$

with α_0 the carrier-level transmission, m the modulation index and ω_m the modulating frequency.

Step 3: check the condition

- Column sum: $|S_{11}|^2 + |S_{21}|^2 = \alpha^2$, which equals 1 only when $\alpha = 1$.
- So the modulator is **never lossless** except at full transmission: the dissipated fraction is $1 - \alpha^2(t)$, and this is unavoidable. An amplitude modulator works by **throwing power away** in step with the modulating signal.
- It is reciprocal and matched, so the only non-ideal property is the loss.

Step 4: evaluate with numbers

- Take $\alpha_0 = 0.5$ and $m = 0.6$, a typical shallow modulation.

Instant	$\alpha(t)$	$[S]$	IL and P_{out} for 1 W in
Peak, $\cos \omega_m t = +1$	$0.5(1 + 0.6) = 0.80$	$\begin{bmatrix} 0 & 0.8 \\ 0.8 & 0 \end{bmatrix}$	$IL = -20 \log 0.8 = 1.94 \text{ dB}; P_{out} = 0.64 \text{ W}$
Carrier, $\cos \omega_m t = 0$	0.5	$\begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix}$	$IL = 6.02 \text{ dB}; P_{out} = 0.25 \text{ W}$
Trough, $\cos \omega_m t = -1$	$0.5(1 - 0.6) = 0.20$	$\begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}$	$IL = 13.98 \text{ dB}; P_{out} = 0.04 \text{ W}$

- Recovering the modulation index from the matrix, which is the check the examiner is looking for:

$$m = \frac{|S_{21}|_{\max} - |S_{21}|_{\min}}{|S_{21}|_{\max} + |S_{21}|_{\min}} = \frac{0.80 - 0.20}{0.80 + 0.20} = \frac{0.60}{1.00} = 0.6 \checkmark$$

- Modulation depth in dB: the transmission swings over $13.98 - 1.94 = 12.04 \text{ dB}$.

Notes worth adding

- **Balanced (reflection-type) realisation:** carrier into the H-arm of a magic tee, identical PIN diodes as reflecting terminations on the two collinear arms, both driven by the modulating bias, and **one diode set back by $\lambda_g/4$** so its round-trip reflection returns 180° shifted, $\Gamma_2 = -\Gamma_1$. A tee with loads Γ_1, Γ_2 gives $S_{44} = (\Gamma_1 + \Gamma_2)/2 = 0$ and $S_{34} = (\Gamma_1 - \Gamma_2)/2 = \Gamma_1$ (Ch4 §4.5, asserted in `wg.py`): the input stays matched and the E-arm output is $\Gamma_1(t)$, which follows the bias. **Without the offset** ($\Gamma_1 = \Gamma_2$) every reflection returns to the H-arm input and nothing reaches the E-arm. verified/added
- S_{11} **is the figure of merit:** a good modulator keeps S_{11} small at every bias point. If it does not, the source sees a varying load and the output picks up incidental phase modulation.
- **Contrast with an ideal switch:** α toggling between 1 and 0 makes the same matrix an on-off (ASK) modulator, with $m = 1$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.